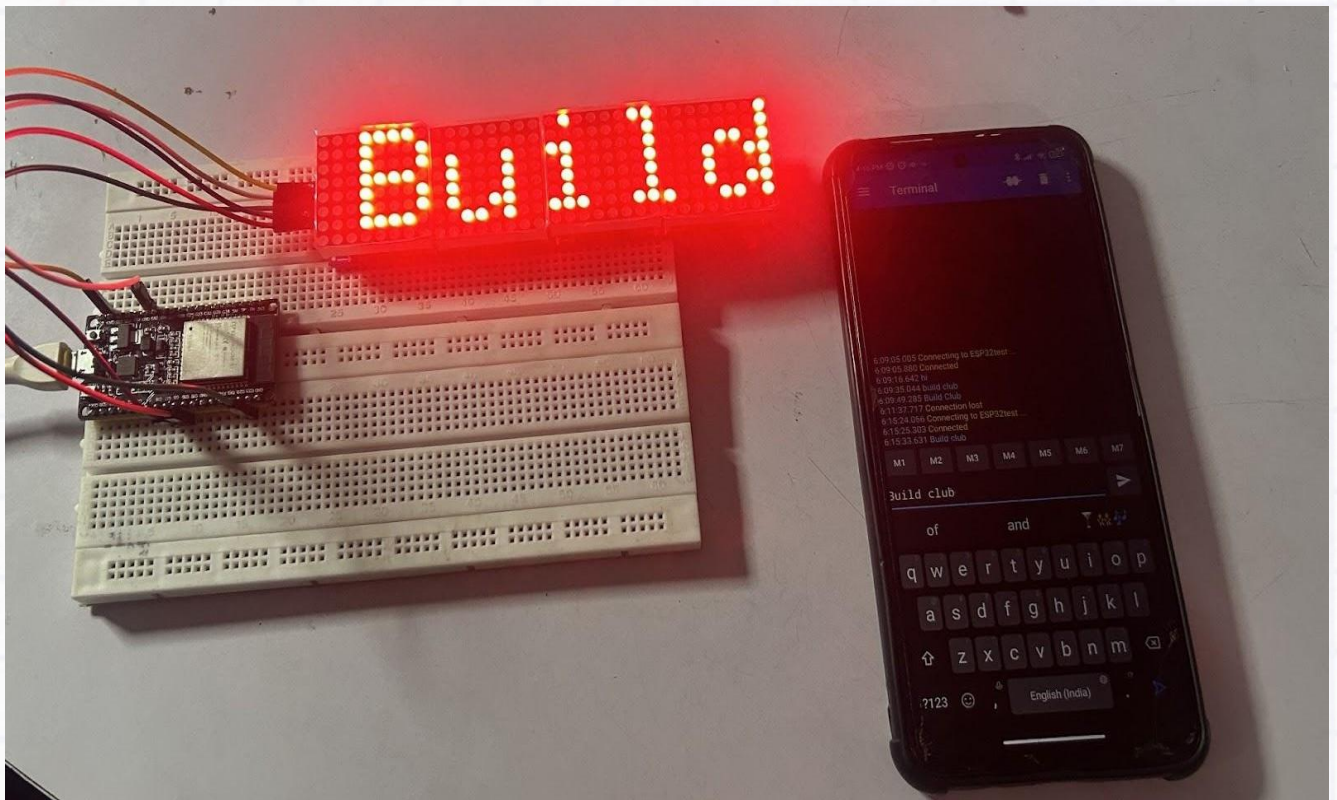


# Build a Rolling LED Display

(A dot-matrix rolling display that scrolls text with adjustable speed and brightness controls)



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## Prerequisites

| Topic                | Resources                                        |
|----------------------|--------------------------------------------------|
| Dancing LEDs Project | Dancing LEDs Project                             |
| Basic C Programming  | <a href="#">C Full Course - Youtube</a>          |
|                      | <a href="#">W3Schools interactive C Tutorial</a> |
|                      | <a href="#">Learn-C.org interactive Tutorial</a> |
| Number Systems       | <a href="#">Binary,Decimal &amp; HexaDecimal</a> |

## Outcome

Build an LED matrix display and send messages to it wirelessly from your phone using Bluetooth. This includes controlling the Direction and Speed of scrolling as well as Brightness of the display remotely from your phone.

## Components

| Sno | Components with specification   | Quantity |
|-----|---------------------------------|----------|
| 1.  | Max 7219 Display panel          | 1        |
| 2.  | ESP32 C6 Dev Module             | 1        |
| 3.  | Jumper wires (Female to Female) | 5        |
| 4.  | USB to Type – C cable           | 1        |



ESP32 C-6 C-6 Module



Dot Matrix Display



Jumper Wires



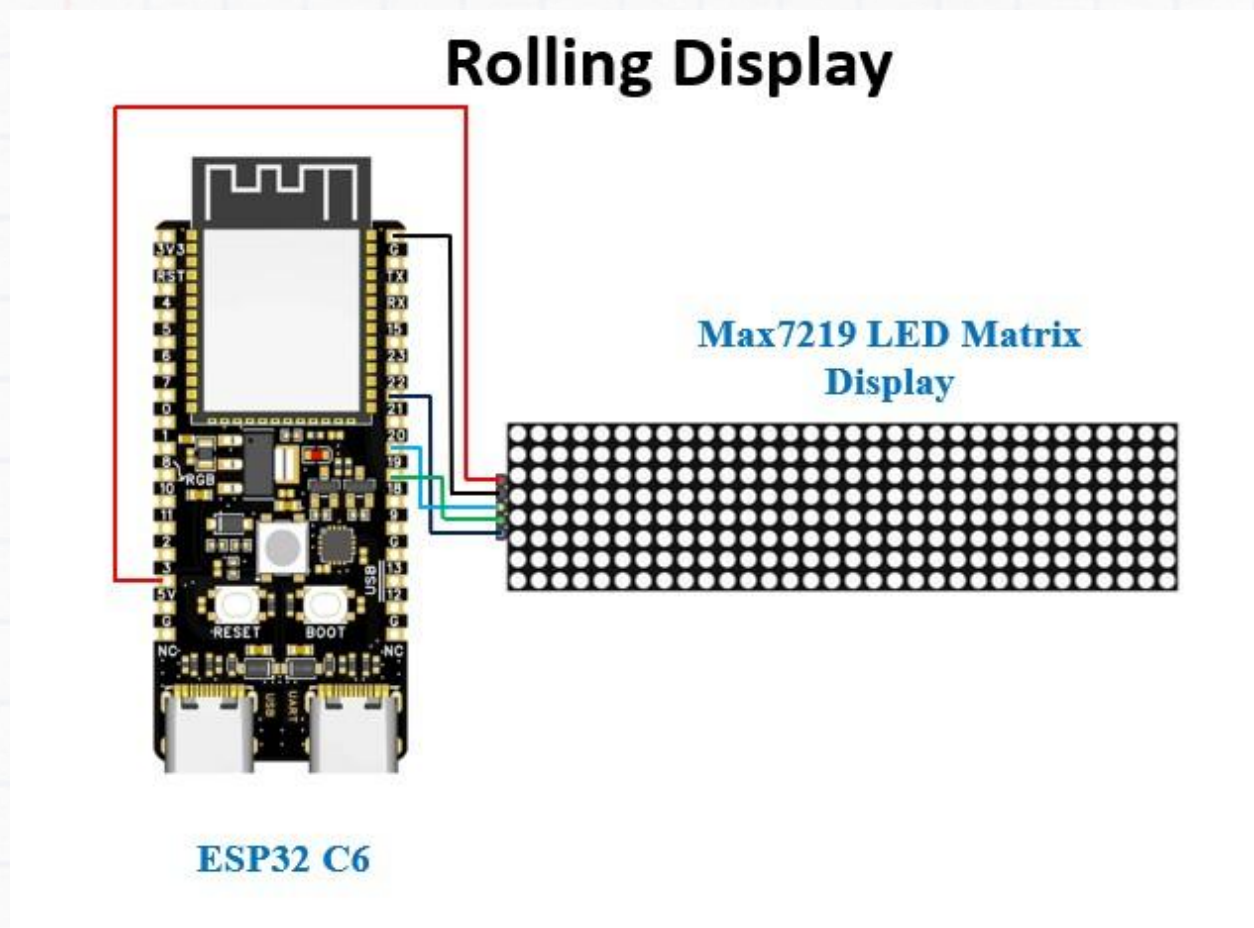
USB



## Connections

Safety tip: Always ensure that the connections to the components are correct and completed before connecting the power supply to the ESP32 C-6.

### Circuit Diagram



Follow the detailed steps in the following pages to complete the circuit.

NOTE: Before starting the connections, verify that all the jumper wires are working using a multimeter. Also ensure that the connections are strong, or else the setup may not work.

#### Detailed Connection Steps

Take 5 female-to-female jumper wires and connect them to the pins (VCC, GND, DIN, CS, CLK) of the MAX7219 display board as shown below. Connect the other ends as per the below connections:

- VCC to 5V of ESP32C-6
- GND to G of ESP32C-6
- DIN to GPIO19 of ESP32 C-6

- CS to GPIO18 of ESP32 C-6
- CLK to GPIO21 of ESP32 C-6

You are now ready to work on the software for the project.

## Software

Launching the IDE for our project

### 1. Install the Arduino IDE

If you haven't already installed the Arduino IDE, download and install it from the official Arduino website.

### 2. Install the ESP32 C-6 Board in Arduino IDE

- Open the Arduino IDE.
- Go to **File > Preferences**.
- In the "Additional Board Manager URLs" field, add the following URL:

[https://raw.githubusercontent.com/espressif/arduino-ESP32 C-6/gh-pages/package\\_ESP32 C-6\\_index.json](https://raw.githubusercontent.com/espressif/arduino-ESP32 C-6/gh-pages/package_ESP32 C-6_index.json)

- Go to **Tools > Board > Boards Manager**.
- Search for ESP32 C-6 and install the ESP32 C-6 package by Espressif Systems.

### 3. Install Required Libraries

- Go to **Sketch > Include Library > Manage Libraries**.

Search for and install the following libraries:

- Please use the [link](#) to Download the library and extract it and paste it Documents > Arduino > Libraries.
- Make sure the 'Libraries' folder located in Documents > Arduino > Libraries.
- BluetoothSerial (if not included by default with the ESP32 C-6 package)

### 4. Connect Your ESP32 C-6 Board

- Connect your ESP32 C-6 board to your computer using a USB cable.
- Go to Tools > Board > ESP32 C-6 Dev Module.
- Go to Tools > Port and select the COM port to which your ESP32 C-6 is connected.

### 5. Prepare the Hardware

Connect your LED matrix to the ESP32 C-6. For example, ensure the CS pin of the matrix is connected to GPIO 19 of the ESP32 C-6.

### 6. Upload the Code

- Copy the provided code below into a new sketch in the Arduino IDE.
- Click the upload button (right arrow) to compile and upload the code to your ESP32 C-6.

#### 7. Pairing the ESP32 C-6 via Bluetooth

- Once the code is uploaded, open the Serial Monitor from **Tools > Serial Monitor** and set the baud rate to 115200.
- Pair your ESP32 C-6 with your computer or smartphone. The device name should be ESP32 C-6test.
- Once paired, you can send text messages to the ESP32 C-6 via a Bluetooth terminal app on your smartphone or a serial terminal on your computer.

Once the code is uploaded to ESP32 C-6, the LED matrix will start displaying the text in various alignments.

### Code to Upload

Ensure you have the following code in your Arduino IDE before uploading:

#### Level 1: Just display the text

For this level, we'll focus on setting up the LED matrix and displaying static text.

- Create a new sketch on arduino IDE and name it displayText.ino.
- Copy the code given below and make some changes according to the input.

Here's the simplified code:

```
#include <SPI.h>
#include <Adafruit_GFX.h>
#include <Max72xxPanel.h>
#include <MD_MAX72xx.h>

const int pinCS = 18;
// CLK - GPIO 21 (For C-6 Board)
// DIN - GPIO 19 (For C-6 Board)
const int numberOfHorizontalDisplays = 4;
const int numberOfVerticalDisplays = 1;

Max72xxPanel matrix = Max72xxPanel(pinCS, numberOfHorizontalDisplays,
numberOfVerticalDisplays);

void setup()
{
  initializeMatrix();
  displayStaticText ("Hello");
```



```

}
void loop()
{
}

void initializeMatrix()
{
matrix.setIntensity(15); // Adjust the brightness between 0 and 15
for (int i = 0; i < numberOfHorizontalDisplays; i++)
{
matrix.setPosition(i, i, 0); // Set positions for all displays. First i is the index, second i is the
horizontal panel and third is the vertical panel
matrix.setRotation(i, 1); // Set rotation for all displays
}
}

void displayStaticText(String text)
{
matrix.fillScreen(LOW);
matrix.setCursor(1, 0);
matrix.print(text);
matrix.write();
}

```

## Level 2: Scroll the text

```

#include <SPI.h>
#include <Adafruit_GFX.h>
#include <Max72xxPanel.h>
#include <MD_MAX72xx.h>

const int pinCS = 18;
const int numberOfHorizontalDisplays = 4;
const int numberOfVerticalDisplays = 1;

//Pin Number 19 - DIN
//Pin Number 21 - CLK

Max72xxPanel matrix = Max72xxPanel(pinCS, numberOfHorizontalDisplays,
numberOfVerticalDisplays);

void setup() {
  initializeMatrix();
}

```

```
void loop() {
    displayScrollingText("Hello");
}

void initializeMatrix()
{
    matrix.setIntensity(1); // Adjust brightness
    for (int i = 0; i < numberOfHorizontalDisplays; i++)
    {
        matrix.setPosition(i, i, 0); // Set panel positions
        matrix.setRotation(i, 1); // Set panel rotation
    }
}

void displayScrollingText(String text)
{
    int textWidth = text.length() * 6; // Each character is 6 pixels wide
    for (int i = numberOfHorizontalDisplays * 8; i > -textWidth; i--)
    // To scroll text from right to left

    {
        matrix.fillScreen(LOW); // Clear the display
        matrix.setCursor(i, 1); // Move text horizontally
        matrix.print(text); // Draw the text
        matrix.write(); // Update the display
        delay(100); // Adjust speed of scrolling
    }
}
```

### Level 3: Add Wireless Control functionality via IP (Internet Protocol)

```
#include <WiFi.h>
#include <MD_Parola.h>
#include <Max72xxPanel.h>
#include <MD_MAX72xx.h>
#include <SPI.h>

// ----- DISPLAY CONFIGURATION -----
#define HARDWARE_TYPE MD_MAX72XX::FC16_HW
#define MAX_DEVICES 4
#define DATA_PIN 19
#define CS_PIN 18
#define CLK_PIN 21
```



```
MD_Parola display = MD_Parola(HARDWARE_TYPE, DATA_PIN, CLK_PIN, CS_PIN,
MAX_DEVICES);

// ----- WIFI CONFIGURATION -----
const char* ssid = "YOUR_WIFI_SSID";
const char* password = "YOUR_WIFI_PASSWORD";
WiFiServer server(80);

// ----- GLOBAL STATE -----
String inputText = "HELLO!";
uint16_t scrollSpeed = 100; // in milliseconds (corrected)
textEffect_t scrollDir = PA_SCROLL_LEFT;
uint8_t brightness = 3;

void setup() {
  Serial.begin(115200);
  display.begin();
  display.setIntensity(brightness);
  display.displayClear();
  display.displayScroll(inputText.c_str(), PA_LEFT, scrollDir, scrollSpeed);

  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi connected: ");
  Serial.println(WiFi.localIP());

  server.begin();
}

void loop() {
  if (display.displayAnimate()) display.displayReset();

  WiFiClient client = server.available();
  if (client) {
    String request = client.readStringUntil('\r');
    client.flush();

    // --- Parse Parameters Safely ---
    if (request.indexOf("GET /?") >= 0) {
      if (request.indexOf("text=") >= 0) {
        int start = request.indexOf("text=") + 5;
        int end = request.indexOf("&", start);
        if (end == -1) end = request.indexOf(" ", start);
        inputText = request.substring(start, end);
      }
    }
  }
}
```

```

    inputText.replace("+", " ");
    inputText.replace("%20", " ");
}

if (request.indexOf("speed=") >= 0) {
    int start = request.indexOf("speed=") + 6;
    int end = request.indexOf("&", start);
    if (end == -1) end = request.indexOf(" ", start);
    scrollSpeed = request.substring(start, end).toInt();
}

if (request.indexOf("bright=") >= 0) {
    int start = request.indexOf("bright=") + 7;
    int end = request.indexOf("&", start);
    if (end == -1) end = request.indexOf(" ", start);
    brightness = constrain(request.substring(start, end).toInt(), 0, 15);
}

if (request.indexOf("dir=") >= 0) {
    int start = request.indexOf("dir=") + 4;
    int end = request.indexOf("&", start);
    if (end == -1) end = request.indexOf(" ", start);
    String dirVal = request.substring(start, end);
    if (dirVal == "right") scrollDir = PA_SCROLL_RIGHT;
    else scrollDir = PA_SCROLL_LEFT;
}

// Update display
display.setIntensity(brightness);
display.displayClear();
display.displayScroll(inputText.c_str(), PA_LEFT, scrollDir, scrollSpeed);
}

// --- HTML Response ---
client.println("HTTP/1.1 200 OK");
client.println("Content-Type: text/html\n");
client.println("<!DOCTYPE html><html><head><title>LED Matrix</title>");
client.println("<meta name='viewport' content='width=device-width, initial-scale=1.0'>");
client.println("<style>");
client.println("body{background:#111;color:#fff;font-family:sans-serif;padding:20px;text-align:center;}");
client.println("input,select{padding:10px;margin:5px;border:none;border-radius:5px;}");
client.println("input[type=range]{width:80%;}");
client.println("button{padding:10px 20px;border:none;background:#09f;color:#fff;border-radius:5px;}");
client.println("</style></head><body>");
client.println("<h2>ESP32 C-6 LED Matrix Control</h2>");

```

```

client.println("<form method='GET'>");

// Text
client.println("Text:<br><input type='text' name='text' value='" + inputText + "'><br>");

// Scroll speed (invert user input logic)
int userSpeed = scrollSpeed;
client.println("Speed (ms):<br><input type='range' min='10' max='300' name='speed'
value='" + String(userSpeed) + "'><br>");

// Brightness
client.println("Brightness:<br><input type='range' min='0' max='15' name='bright'
value='" + String(brightness) + "'><br>");

// Direction
client.println("Direction:<br><select name='dir'>");
client.println("<option value='left'" + String(scrollDir == PA_SCROLL_LEFT ? " selected"
: "")) + ">Left</option>");
client.println("<option value='right'" + String(scrollDir == PA_SCROLL_RIGHT ? "
selected" : "")) + ">Right</option>");
client.println("</select><br><br>");

client.println("<button type='submit'>Update</button>");
client.println("</form></body></html>");

delay(10);
client.stop();
}
}

```





### Workshop Tasks:

- 1) Write code to print digits 0-9 on the display. Try similarly for punctuation marks '!', '?', ','.
- 2) Design 3 emojis or symbols of your choice and print them on the display.
- 3) Write a program that animates a 'Stick man' walking from one end of the display to the other.
- 4) Plan an animation clip of minimum 10 seconds with a storyline and run it on the display.

### Club Activity:

- 1) Write code to execute two different text messages can be print on the display using timer function.
- 2) Write a code which can executes the text messages on the display using WIFI.
- 3) Write a code which can store all the text messages printed on the display using google sheets.